

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology**  
**Practical - XV**

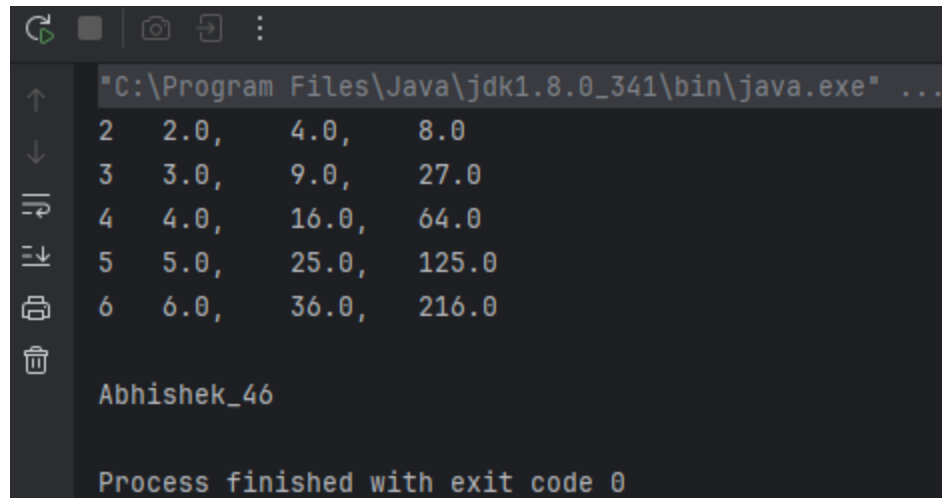
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|------------------------------|-----------------------------------|
| Roll No.:S046                | Name: Abhishek Yadav              |
| Class: BSC IT                | Batch:-----                       |
| Date of Assignment:11/8/2026 | Date/Time of Submission:11/8/2026 |

**Aim:** Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.

**Code:**

```
build.sbt  Demo15.scala x
1  object Demo15 extends App {
2
3      val data = List(2, 3, 4, 5, 6)
4
5      for (x <- data) {
6          val p1 = math.pow(x, 1)
7          val p2 = math.pow(x, 2)
8          val p3 = math.pow(x, 3)
9
10         println(s"$x\t$p1, \t$p2, \t$p3")
11     }
12     println("\nAbhishek_46")
13 }
```

**Output:**



The image shows a screenshot of an IDE's console window. The title bar at the top indicates the process is running `"C:\Program Files\Java\jdk1.8.0_341\bin\java.exe" ...`. On the left side of the console, there is a vertical toolbar with icons for: up arrow, down arrow, undo, redo, copy, and delete. The main area of the console displays the following output:

```
2  2.0,    4.0,    8.0
3  3.0,    9.0,   27.0
4  4.0,   16.0,   64.0
5  5.0,   25.0,  125.0
6  6.0,   36.0,  216.0
```

Below the numerical output, the text `Abhishek_46` is printed. At the bottom of the console, a status message reads: `Process finished with exit code 0`.